In [119…

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
sns.set(style="whitegrid")
```

importing data

In [2]:

```
df = pd.read_csv('Hotel Reservations.csv')
df
```

Out[2]:

	Booking_ID	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	type_of_meal_plan	required_car_parking_space
0	INN00001	2	0	1	2	Meal Plan 1	0
1	INN00002	2	0	2	3	Not Selected	0
2	INN00003	1	0	2	1	Meal Plan 1	0
3	INN00004	2	0	0	2	Meal Plan 1	0
4	INN00005	2	0	1	1	Not Selected	0
...	...	...	...	...	...	...	...
36270	INN36271	3	0	2	6	Meal Plan 1	0
36271	INN36272	2	0	1	3	Meal Plan 1	0
36272	INN36273	2	0	2	6	Meal Plan 1	0
36273	INN36274	2	0	0	3	Not Selected	0
36274	INN36275	2	0	1	2	Meal Plan 1	0

36275 rows × 19 columns

Data Cleaning

In [3]:

```
df.head(5)
```

Out[3]:

	Booking_ID	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	type_of_meal_plan	required_car_parking_space	roo
0	INN00001	2	0	1	2	Meal Plan 1	0	
1	INN00002	2	0	2	3	Not Selected	0	
2	INN00003	1	0	2	1	Meal Plan 1	0	
3	INN00004	2	0	0	2	Meal Plan 1	0	
4	INN00005	2	0	1	1	Not Selected	0	

In [4]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 36275 entries, 0 to 36274
Data columns (total 19 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   Booking_ID                            36275 non-null  object
 1   no_of_adults                          36275 non-null  int64
 2   no_of_children                        36275 non-null  int64
 3   no_of_weekend_nights                  36275 non-null  int64
 4   no_of_week_nights                     36275 non-null  int64
 5   type_of_meal_plan                     36275 non-null  object
 6   required_car_parking_space            36275 non-null  int64
 7   room_type_reserved                    36275 non-null  object
 8   lead_time                             36275 non-null  int64
 9   arrival_year                          36275 non-null  int64
 10  arrival_month                         36275 non-null  int64
 11  arrival_date                          36275 non-null  int64
 12  market_segment_type                   36275 non-null  object
 13  repeated_guest                        36275 non-null  int64
 14  no_of_previous_cancellations          36275 non-null  int64
 15  no_of_previous_bookings_not_canceled  36275 non-null  int64
 16  avg_price_per_room                    36275 non-null  float64
 17  no_of_special_requests                36275 non-null  int64
 18  booking_status                        36275 non-null  object
dtypes: float64(1), int64(13), object(5)
memory usage: 5.3+ MB
```

In [5]:
```
df.describe().round(1)
```

Out[5]:

	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	required_car_parking_space	lead_time	arrival_year	arrival_
count	36275.0	36275.0	36275.0	36275.0	36275.0	36275.0	36275.0	3
mean	1.8	0.1	0.8	2.2	0.0	85.2	2017.8	
std	0.5	0.4	0.9	1.4	0.2	85.9	0.4	
min	0.0	0.0	0.0	0.0	0.0	0.0	2017.0	
25%	2.0	0.0	0.0	1.0	0.0	17.0	2018.0	
50%	2.0	0.0	1.0	2.0	0.0	57.0	2018.0	
75%	2.0	0.0	2.0	3.0	0.0	126.0	2018.0	
max	4.0	10.0	7.0	17.0	1.0	443.0	2018.0	

In [6]:
```
df.isna().sum()
```

Out[6]:
```
Booking_ID                            0
no_of_adults                          0
no_of_children                        0
no_of_weekend_nights                  0
no_of_week_nights                     0
type_of_meal_plan                     0
required_car_parking_space            0
room_type_reserved                    0
lead_time                             0
arrival_year                          0
arrival_month                         0
arrival_date                          0
market_segment_type                   0
repeated_guest                        0
no_of_previous_cancellations          0
no_of_previous_bookings_not_canceled  0
avg_price_per_room                    0
no_of_special_requests                0
booking_status                        0
dtype: int64
```

In [7]:
```
df.duplicated().sum()
```

Out[7]:
```
np.int64(0)
```

In [8]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 36275 entries, 0 to 36274
Data columns (total 19 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   Booking_ID                            36275 non-null  object
 1   no_of_adults                          36275 non-null  int64
 2   no_of_children                        36275 non-null  int64
 3   no_of_weekend_nights                  36275 non-null  int64
 4   no_of_week_nights                     36275 non-null  int64
 5   type_of_meal_plan                     36275 non-null  object
 6   required_car_parking_space            36275 non-null  int64
 7   room_type_reserved                    36275 non-null  object
 8   lead_time                             36275 non-null  int64
 9   arrival_year                          36275 non-null  int64
 10  arrival_month                         36275 non-null  int64
 11  arrival_date                          36275 non-null  int64
 12  market_segment_type                   36275 non-null  object
 13  repeated_guest                        36275 non-null  int64
 14  no_of_previous_cancellations          36275 non-null  int64
 15  no_of_previous_bookings_not_canceled  36275 non-null  int64
 16  avg_price_per_room                    36275 non-null  float64
 17  no_of_special_requests                36275 non-null  int64
 18  booking_status                        36275 non-null  object
dtypes: float64(1), int64(13), object(5)
memory usage: 5.3+ MB
```

In [9]:
```
df['type_of_meal_plan'].value_counts()
```

Out[9]:
```
type_of_meal_plan
Meal Plan 1     27835
Not Selected     5130
Meal Plan 2      3305
Meal Plan 3         5
Name: count, dtype: int64
```

In [10]:
```
df['required_car_parking_space'] = df['required_car_parking_space'].replace(1,'Yes').replace(0,'No')
```

In [11]:
```
df['required_car_parking_space'].value_counts()
```

Out[11]:
```
required_car_parking_space
No     35151
Yes     1124
Name: count, dtype: int64
```

In [12]:
```
df.rename(columns= {'arrival_year': 'year' , 'arrival_month':'month' , 'arrival_date':'day'},inplace= True)
```

In [13]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 36275 entries, 0 to 36274
Data columns (total 19 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   Booking_ID                            36275 non-null  object
 1   no_of_adults                          36275 non-null  int64
 2   no_of_children                        36275 non-null  int64
 3   no_of_weekend_nights                  36275 non-null  int64
 4   no_of_week_nights                     36275 non-null  int64
 5   type_of_meal_plan                     36275 non-null  object
 6   required_car_parking_space            36275 non-null  object
 7   room_type_reserved                    36275 non-null  object
 8   lead_time                             36275 non-null  int64
 9   year                                  36275 non-null  int64
 10  month                                 36275 non-null  int64
 11  day                                   36275 non-null  int64
 12  market_segment_type                   36275 non-null  object
 13  repeated_guest                        36275 non-null  int64
 14  no_of_previous_cancellations          36275 non-null  int64
 15  no_of_previous_bookings_not_canceled  36275 non-null  int64
 16  avg_price_per_room                    36275 non-null  float64
 17  no_of_special_requests                36275 non-null  int64
 18  booking_status                        36275 non-null  object
dtypes: float64(1), int64(12), object(6)
memory usage: 5.3+ MB
```

In [14]:
```
df['date'] = pd.to_datetime(df[['year','month','day']] , errors='coerce')
```

In [15]:
```
df['date'].isna().sum()
```

Out[15]:
```
np.int64(37)
```

In [16]:
```
df[df['date'].isna()]
```

Out[16]:

	Booking_ID	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	type_of_meal_plan	required_car_parking_space
2626	INN02627	2	0	1	5	Meal Plan 1	No
3677	INN03678	1	0	1	3	Meal Plan 1	No
5600	INN05601	2	0	1	3	Meal Plan 1	No
6343	INN06344	1	0	1	1	Meal Plan 1	No
7648	INN07649	2	1	1	5	Meal Plan 1	No
8000	INN08001	2	2	1	3	Meal Plan 1	No
8989	INN08990	1	0	1	2	Meal Plan 1	No
9153	INN09154	2	2	1	3	Meal Plan 1	No
9245	INN09246	2	0	1	3	Meal Plan 1	No
9664	INN09665	1	0	1	0	Meal Plan 1	No
9934	INN09935	1	0	1	2	Meal Plan 1	No
10593	INN10594	2	0	1	3	Meal Plan 1	Yes
10652	INN10653	2	0	1	2	Meal Plan 1	No
10747	INN10748	2	0	1	3	Meal Plan 1	No
11881	INN11882	1	0	3	7	Meal Plan 1	No
13958	INN13959	1	0	1	1	Meal Plan 1	No
14304	INN14305	2	0	1	3	Meal Plan 2	No
15363	INN15364	1	0	1	2	Meal Plan 1	No
15438	INN15439	1	0	1	2	Meal Plan 1	No
17202	INN17203	2	0	1	3	Meal Plan 2	No
18380	INN18381	1	0	1	1	Meal Plan 1	No
18534	INN18535	1	0	1	2	Meal Plan 1	No
18680	INN18681	1	0	1	4	Meal Plan 1	Yes
19013	INN19014	1	0	1	1	Meal Plan 1	Yes
20419	INN20420	2	0	1	1	Meal Plan 1	No
21674	INN21675	1	0	1	1	Meal Plan 1	No
21688	INN21689	2	0	1	2	Meal Plan 1	No
26108	INN26109	2	0	1	2	Meal Plan 1	No
27559	INN27560	1	0	1	0	Meal Plan 1	No
27928	INN27929	2	0	1	5	Meal Plan 1	No
30552	INN30553	2	1	1	3	Meal Plan 1	No
30616	INN30617	1	0	1	0	Meal Plan 1	No
30632	INN30633	3	0	1	2	Meal Plan 2	No
30839	INN30840	1	0	1	2	Meal Plan 1	No
32041	INN32042	2	0	1	0	Not Selected	No
34638	INN34639	1	0	1	2	Meal Plan 1	No
35481	INN35482	1	0	1	1	Meal Plan 1	No

In [17]:
```
df[df['date'].isna()].tail()
```

Out[17]:

	Booking_ID	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	type_of_meal_plan	required_car_parking_space
30632	INN30633	3	0	1	2	Meal Plan 2	No
30839	INN30840	1	0	1	2	Meal Plan 1	No
32041	INN32042	2	0	1	0	Not Selected	No
34638	INN34639	1	0	1	2	Meal Plan 1	No
35481	INN35482	1	0	1	1	Meal Plan 1	No

In [18]:
```
# يوم 29 شهر 2 والسنه مش كبيسه كده التاريخ دا غلط نعدله ل 28
```

In [19]:
```
mask = (df['month'] == 2) & (df['day'] == 29)
df.loc[mask, 'day'] = 28
```

In [20]:
```
df = df.drop(columns=['date'])
```

In [21]:
```
df['date'] = pd.to_datetime(df[['year','month','day']] , errors='coerce')
```

In [22]:
```
df['date'].isna().sum()
```

Out[22]:
```
np.int64(0)
```

In [23]:
```
df['repeated_guest'].value_counts()
```

Out[23]:
```
repeated_guest
0    35345
1      930
Name: count, dtype: int64
```

In [24]:
```
df['repeated_guest'] = df['repeated_guest'].replace(0,'No').replace(1,'Yes')
```

In [25]:
```
df['repeated_guest'].value_counts()
```

Out[25]:
```
repeated_guest
No     35345
Yes      930
Name: count, dtype: int64
```

In [26]:
```
df['booking_status'].value_counts()
```

Out[26]:
```
booking_status
Not_Canceled    24390
Canceled        11885
Name: count, dtype: int64
```

In [27]:
```
df
```

Out[27]:

	Booking_ID	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	type_of_meal_plan	required_car_parking_space
0	INN00001	2	0	1	2	Meal Plan 1	No
1	INN00002	2	0	2	3	Not Selected	No
2	INN00003	1	0	2	1	Meal Plan 1	No
3	INN00004	2	0	0	2	Meal Plan 1	No
4	INN00005	2	0	1	1	Not Selected	No
...	...	...	...	...	...	...	...
36270	INN36271	3	0	2	6	Meal Plan 1	No
36271	INN36272	2	0	1	3	Meal Plan 1	No
36272	INN36273	2	0	2	6	Meal Plan 1	No
36273	INN36274	2	0	0	3	Not Selected	No
36274	INN36275	2	0	1	2	Meal Plan 1	No

36275 rows × 20 columns

In [ ]:

EDA

In [29]:
```
num_adults = sum(df['no_of_adults'])
num_adults
```

Out[29]:
```
66926
```

In [30]:
```
num_children = sum(df['no_of_children'])
num_children
```

Out[30]: 3819

In [35]:
```
adults_with_childrens = df[df['no_of_children']>0]['no_of_adults'].sum()
adults_with_childrens
```

Out[35]: np.int64(5000)

Type of male distribution

In [36]:
```
df.groupby('type_of_meal_plan')['Booking_ID'].count()
```

Out[36]:
```
type_of_meal_plan
Meal Plan 1     27835
Meal Plan 2      3305
Meal Plan 3         5
Not Selected     5130
Name: Booking_ID, dtype: int64
```

In [49]:
```
df.groupby('type_of_meal_plan').agg({'no_of_weekend_nights':'sum' ,'no_of_week_nights':'sum'})
```

Out[49]:

type_of_meal_plan	no_of_weekend_nights	no_of_week_nights
Meal Plan 1	23158	63261
Meal Plan 2	2246	6668
Meal Plan 3	1	8
Not Selected	4004	10024

Number of guests who required car parking

In [50]:
```
df.groupby('required_car_parking_space')['Booking_ID'].count()
```

Out[50]:
```
required_car_parking_space
No     35151
Yes     1124
Name: Booking_ID, dtype: int64
```

Analysis of Room Types: Popularity, Pricing, and Booking Lead Time

In [58]:
```
room_analysis = df.groupby('room_type_reserved').agg({
    'Booking_ID':'count',
    'lead_time':'mean',
    'avg_price_per_room':'mean'
})
room_analysis.columns = ['Booking_num' , 'AVG_lead_time ' , 'AVG_Price']
room_analysis.round(2).sort_values('AVG_Price' , ascending= False)
```

Out[58]:

room_type_reserved	Booking_num	AVG_lead_time	AVG_Price
Room_Type 6	966	61.02	182.21
Room_Type 7	158	37.09	155.20
Room_Type 4	6057	69.21	125.29
Room_Type 5	265	59.55	123.73
Room_Type 1	28130	89.64	95.92
Room_Type 2	692	101.02	87.85
Room_Type 3	7	69.57	73.68

Yearly Sales Performance: Total Bookings and Revenue per Year

In [60]:
```
df.groupby('year').agg({
    'Booking_ID':'count',
    'avg_price_per_room':'sum'
})
```

Out[60]:

year	Booking_ID	avg_price_per_room
2017	6514	586441.73
2018	29761	3165247.15

Market Segment Performance : Total Booking and Revenue per market

In [63]:
```
df.groupby('market_segment_type').agg({
    'Booking_ID':'count',
    'avg_price_per_room':'sum'
}).sort_values('Booking_ID',ascending=False)
```

Out[63]:

market_segment_type	Booking_ID	avg_price_per_room
Online	23214	2605930.63
Offline	10528	964708.84
Corporate	2017	167232.98
Complementary	391	1228.43
Aviation	125	12588.00

Booking Performance

In [71]:
```
df.groupby('booking_status')['Booking_ID'].count()
```

Out[71]:
```
booking_status
Canceled        11885
Not_Canceled    24390
Name: Booking_ID, dtype: int64
```

Repeated Guest Analysis: Booking Patterns and Cancellation Rates (New vs. Returning)

In [79]:
```
df.groupby('repeated_guest').agg({
    'Booking_ID':'count',
    'no_of_previous_cancellations':'sum',
    'no_of_previous_bookings_not_canceled':'sum',
    'no_of_special_requests':'sum'
})
```

Out[79]:

repeated_guest	Booking_ID	no_of_previous_cancellations	no_of_previous_bookings_not_canceled	no_of_special_requests
No	35345	0	0	21955
Yes	930	847	5565	523

Loyalty Metrics: Past Cancellations vs. Successful Stays for Current Guests

In [81]:
```
confirmed_bookings = df[df['booking_status'] == 'Not_Canceled']
loyalty_analysis = confirmed_bookings.groupby('repeated_guest').agg({
    'Booking_ID':'count',
    'no_of_previous_cancellations':'sum',
    'no_of_previous_bookings_not_canceled':'sum',
    'no_of_special_requests':'sum'
})
loyalty_analysis.columns = ['NUM OF Booking' , 'Cancellations Booking' , 'Confirmed Booking','NUM OF Special Requests' ]
loyalty_analysis
```

Out[81]:

repeated_guest	NUM OF Booking	Cancellations Booking	Confirmed Booking	NUM OF Special Requests
No	23476	0	0	17978
Yes	914	781	5539	523

In [82]:
```
df.loc[:, 'is_canceled'] = df['booking_status'].apply(lambda x: 1 if x == 'Canceled' else 0)
```

In [83]:
```
df.loc[:, 'total_stay_nights'] = df['no_of_week_nights'] + df['no_of_weekend_nights']
```

In [84]:
```
df.loc[:, 'revenue'] = df['total_stay_nights'] * df['avg_price_per_room']
```

Cancellation Rate by Market Segment

In [92]:
```
(df.groupby('market_segment_type')['is_canceled'].mean()*100).round(2).sort_values(ascending=False)
```

Out[92]:
```
market_segment_type
Online           36.51
Offline          29.95
Aviation         29.60
Corporate        10.91
Complementary     0.00
Name: is_canceled, dtype: float64
```

Calncellation Rate by month

In [93]:
```
(df.groupby('month')['is_canceled'].mean()*100).round(2)
```

Out[93]:
```
month
1      2.37
2     25.23
3     29.69
4     36.37
5     36.49
6     40.31
7     45.00
8     39.02
9     33.36
10    35.36
11    29.36
12    13.31
Name: is_canceled, dtype: float64
```

Revenue Per month

In [96]:
```
df.groupby('month')['revenue'].sum().round(1).sort_values(ascending = False)
```

Out[96]:
```
month
10    1605314.1
9     1557440.0
8     1383366.6
7     1073278.2
6      990302.4
12     902368.4
5      874478.4
4      835756.9
11     828055.2
3      684619.9
2      401315.1
1      210983.5
Name: revenue, dtype: float64
```

Cancellation Rate Per Special Requests

In [97]:
```
(df.groupby('no_of_special_requests')['is_canceled'].mean()*100).round(2)
```

Out[97]:
```
no_of_special_requests
0    43.21
1    23.77
2    14.60
3     0.00
4     0.00
5     0.00
Name: is_canceled, dtype: float64
```

Avg lead Time per Booking Status

In [101…
```
df.groupby('booking_status')['lead_time'].mean().round(2)
```

Out[101…
```
booking_status
Canceled        139.22
Not_Canceled     58.93
Name: lead_time, dtype: float64
```

Total Revenue by Room Type for Confirmed Bookings

In [112…
```
confirmed = df[df['booking_status'] == 'Not_Canceled']
room_revenue = confirmed.groupby('room_type_reserved')['revenue'].sum().sort_values(ascending = False)
room_revenue
```

Out[112…

```
room_type_reserved
Room_Type 1    4934054.51
Room_Type 4    1593929.60
Room_Type 6     282425.77
Room_Type 2     122324.28
Room_Type 5      63785.63
Room_Type 7      53469.88
Room_Type 3       1115.00
Name: revenue, dtype: float64
```

Comparative Analysis: Online Segment vs. Other Market Segments (Confirmed Booking)

In [117…

```
confirmed.groupby('market_segment_type').agg({
    'revenue':'sum',
    'lead_time':'mean'
}).round(2)
```

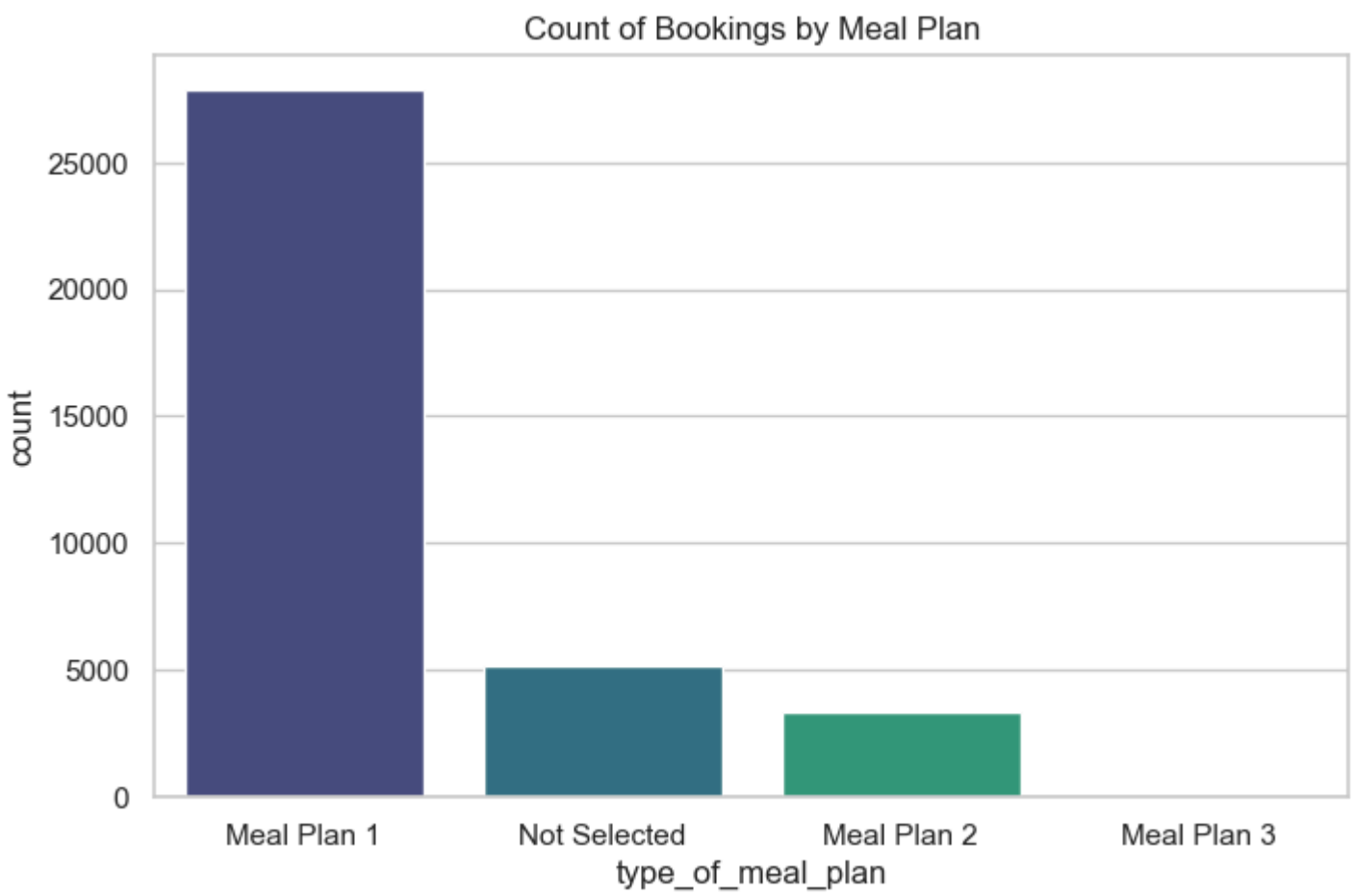
Out[117…

market_segment_type	revenue	lead_time
Aviation	25240.00	4.60
Complementary	2778.99	12.04
Corporate	275493.91	17.66
Offline	1855433.18	85.75
Online	4892158.59	52.11

Data Visualization

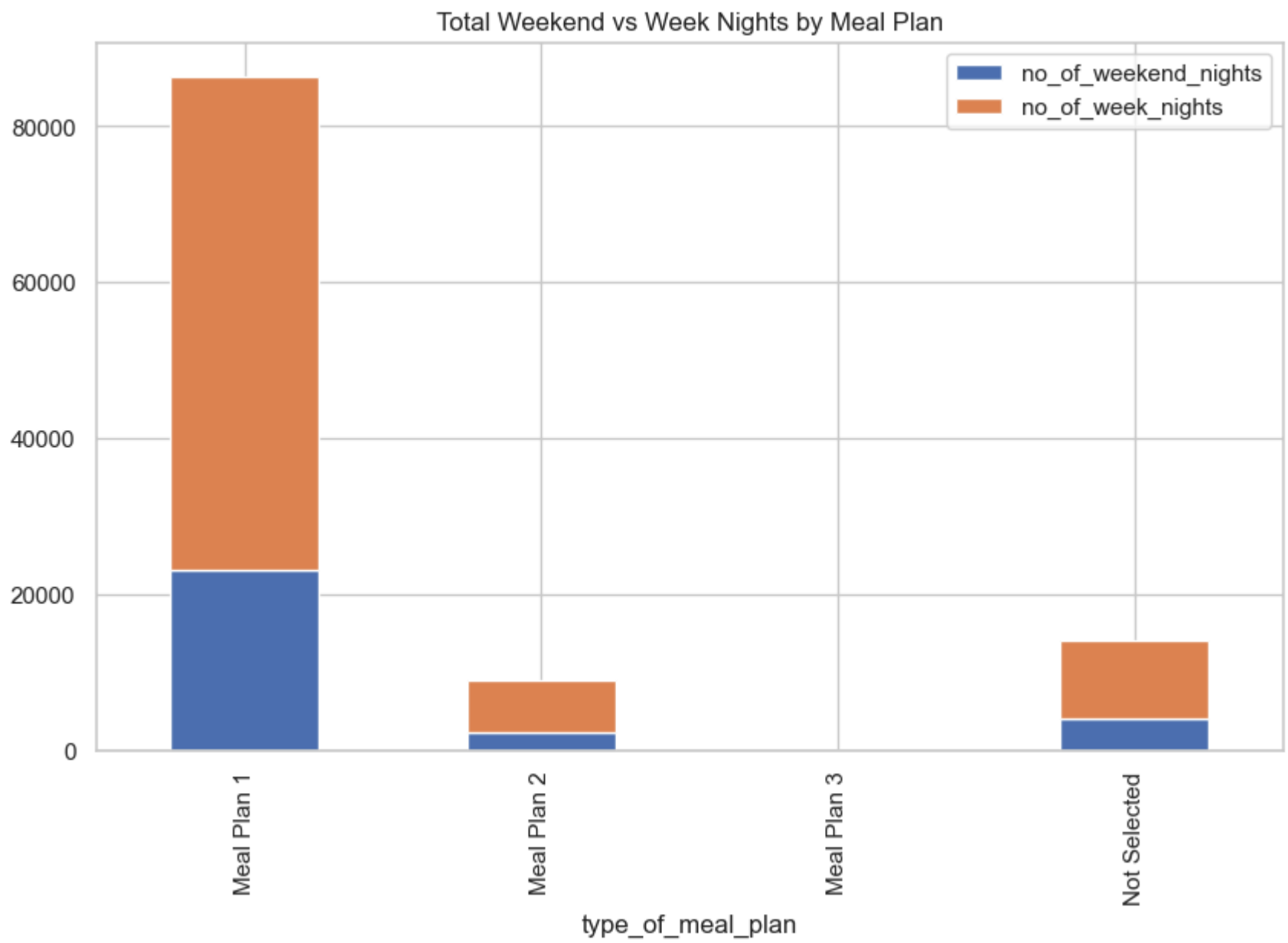
In [121…

```
plt.figure(figsize=(8,5))
sns.countplot(data=df, x='type_of_meal_plan', hue='type_of_meal_plan', palette='viridis', legend=False)
plt.title('Count of Bookings by Meal Plan')
plt.show()
```



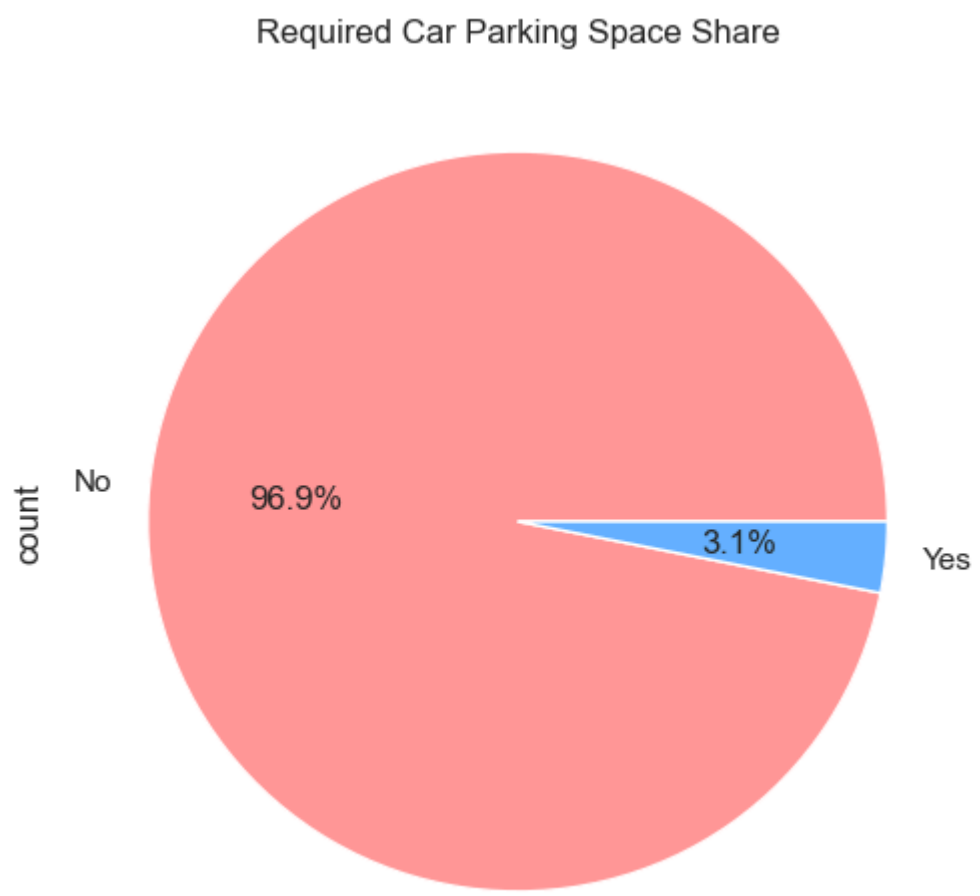
In [122…

```
meal_nights = df.groupby('type_of_meal_plan').agg({'no_of_weekend_nights':'sum' ,'no_of_week_nights':'sum'})
meal_nights.plot(kind='bar', stacked=True, figsize=(10,6))
plt.title('Total Weekend vs Week Nights by Meal Plan')
plt.show()
```



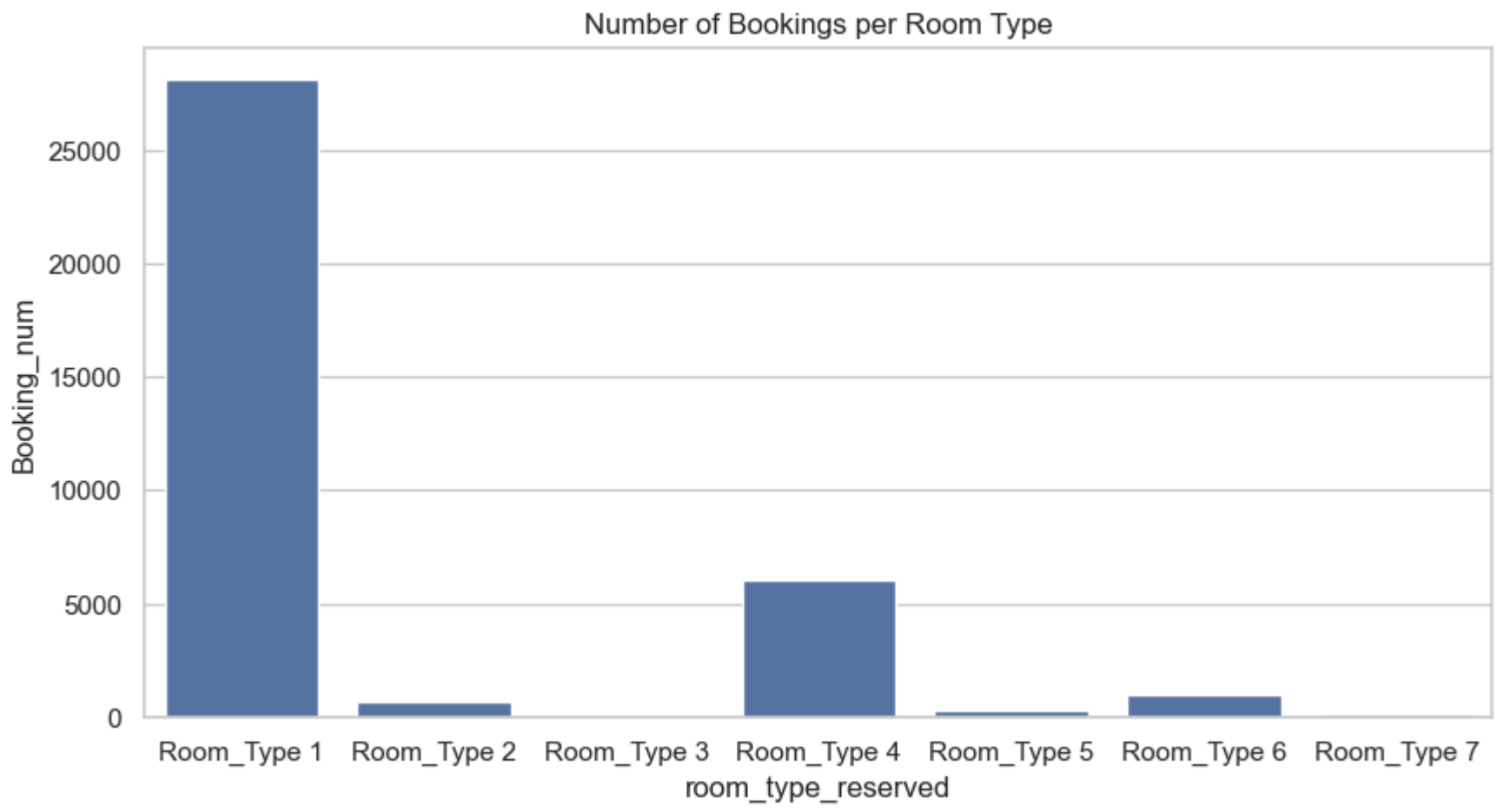
In [123…

```
plt.figure(figsize=(6,6))
df['required_car_parking_space'].value_counts().plot.pie(autopct='%1.1f%%', colors=['#ff9999','#66b3ff'])
plt.title('Required Car Parking Space Share')
plt.show()
```



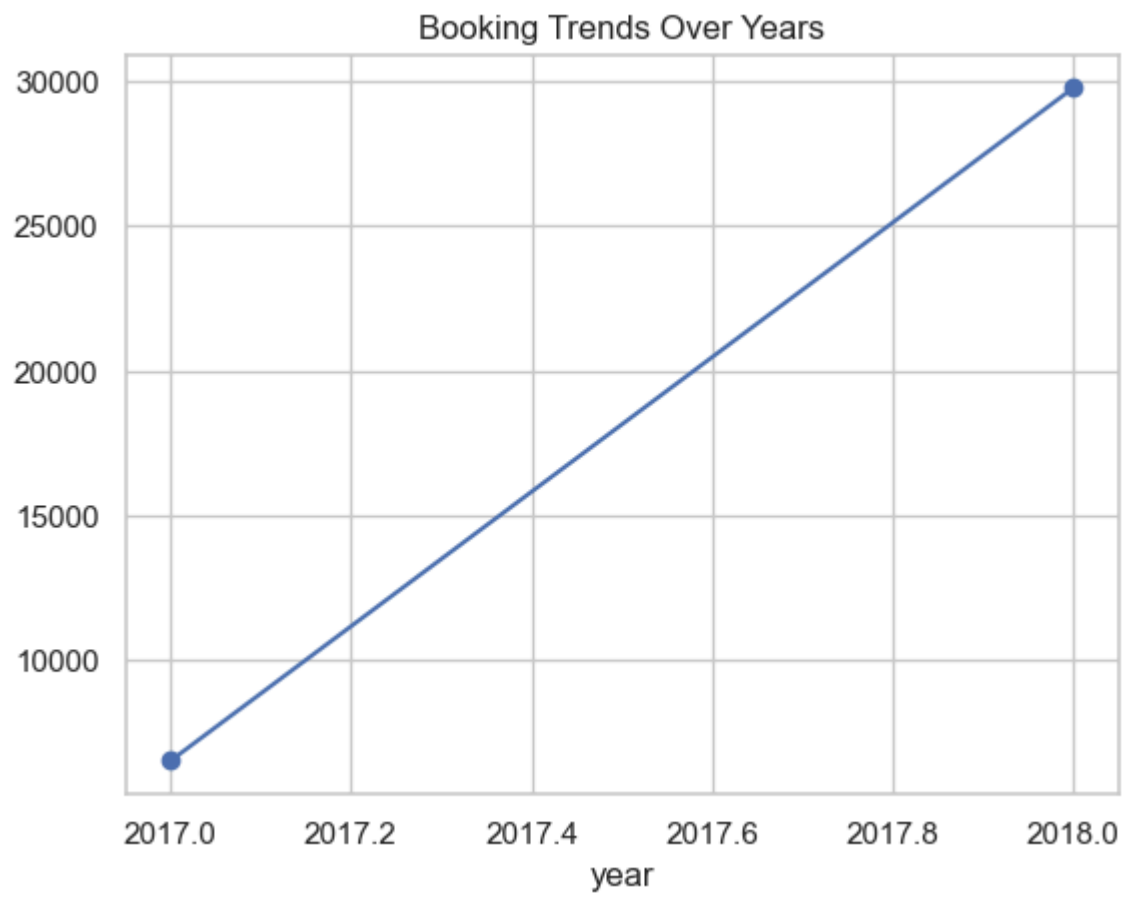
In [124…

```
plt.figure(figsize=(10,5))
sns.barplot(x=room_analysis.index, y='Booking_num', data=room_analysis)
plt.title('Number of Bookings per Room Type')
plt.show()
```



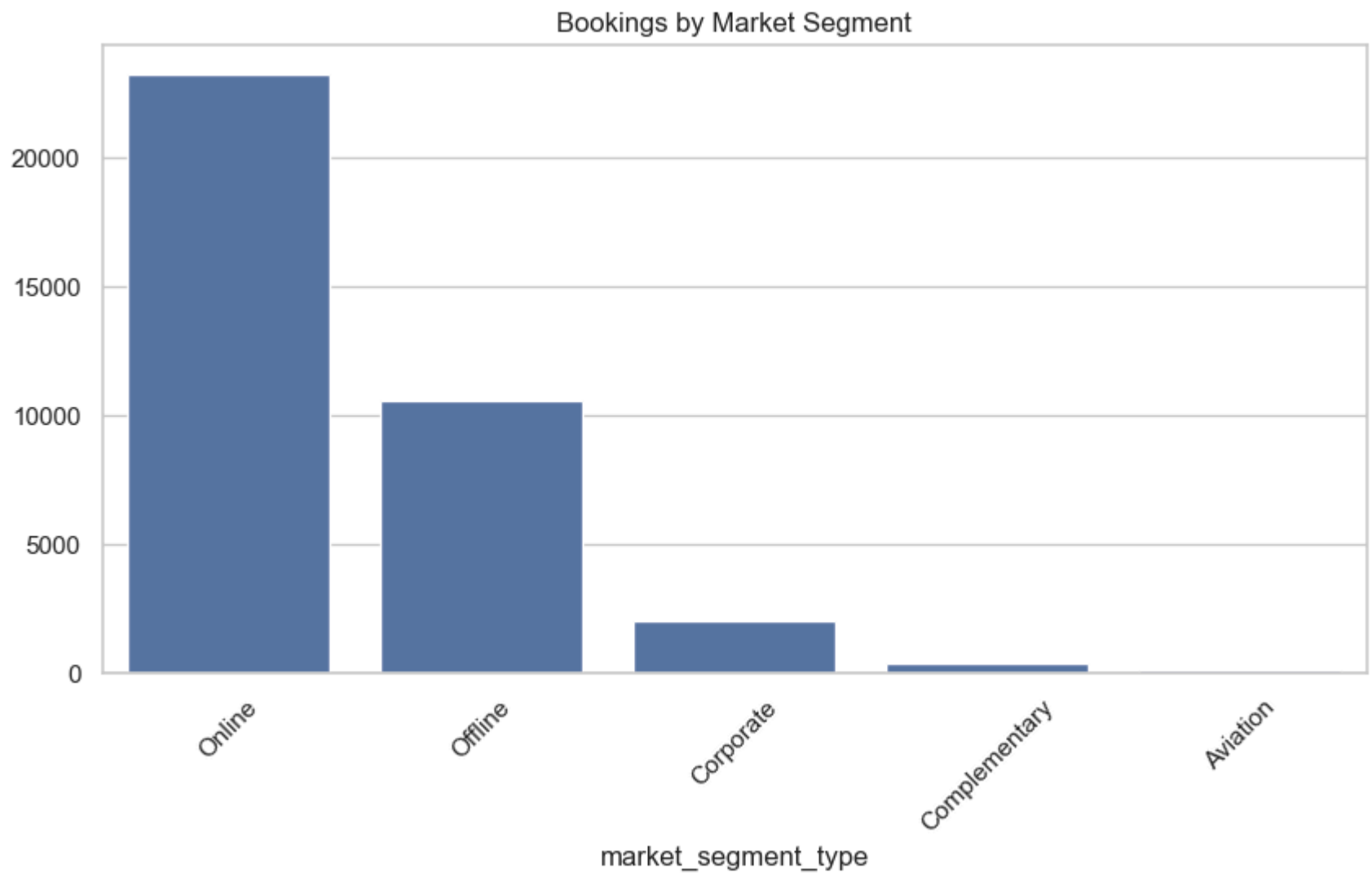
In [125…

```
year_data = df.groupby('year').agg({'Booking_ID':'count', 'avg_price_per_room':'sum'})
year_data['Booking_ID'].plot(kind='line', marker='o', title='Booking Trends Over Years')
plt.show()
```



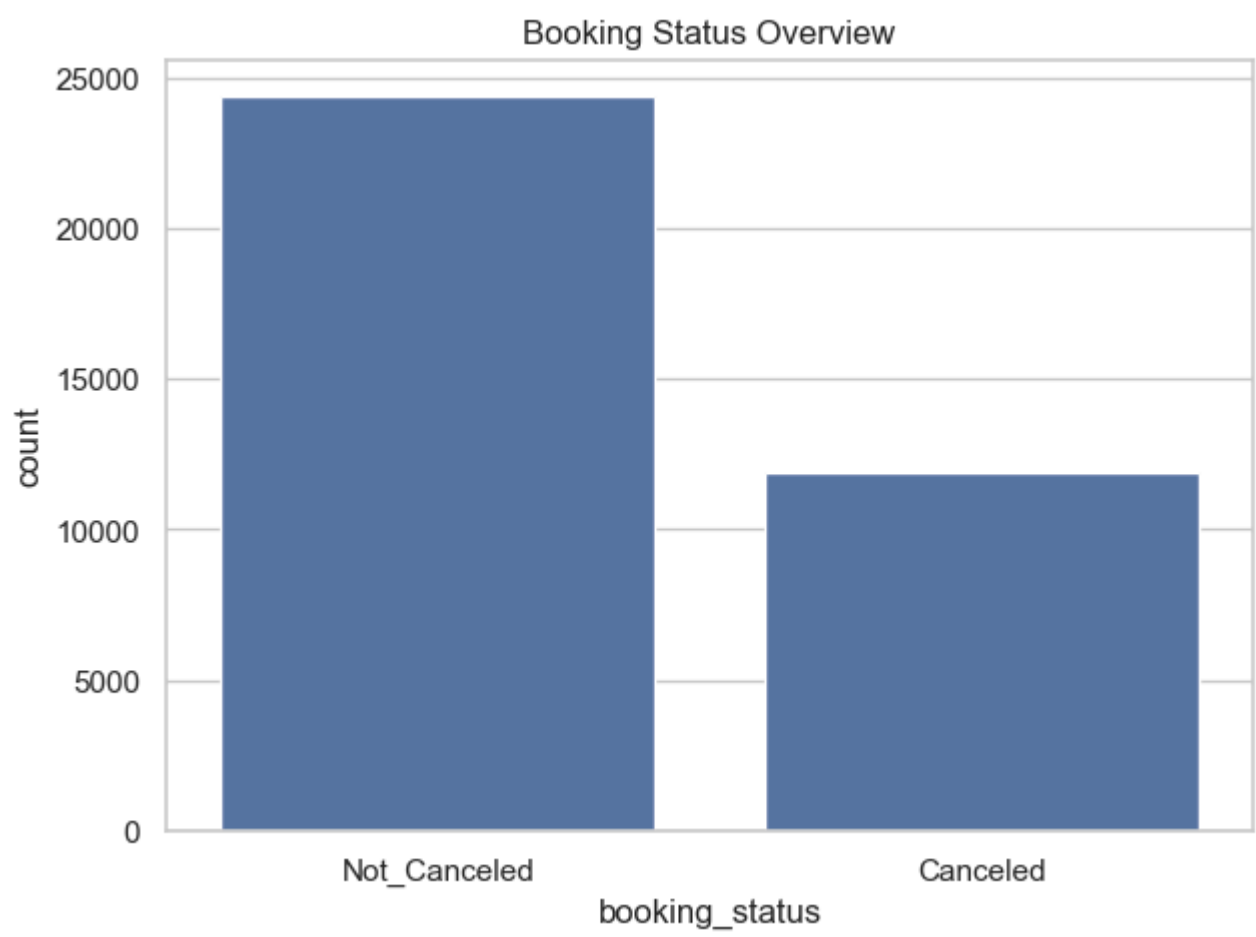
In [126…

```
market_data = df.groupby('market_segment_type')['Booking_ID'].count().sort_values(ascending=False)
plt.figure(figsize=(10,5))
sns.barplot(x=market_data.index, y=market_data.values)
plt.xticks(rotation=45)
plt.title('Bookings by Market Segment')
plt.show()
```



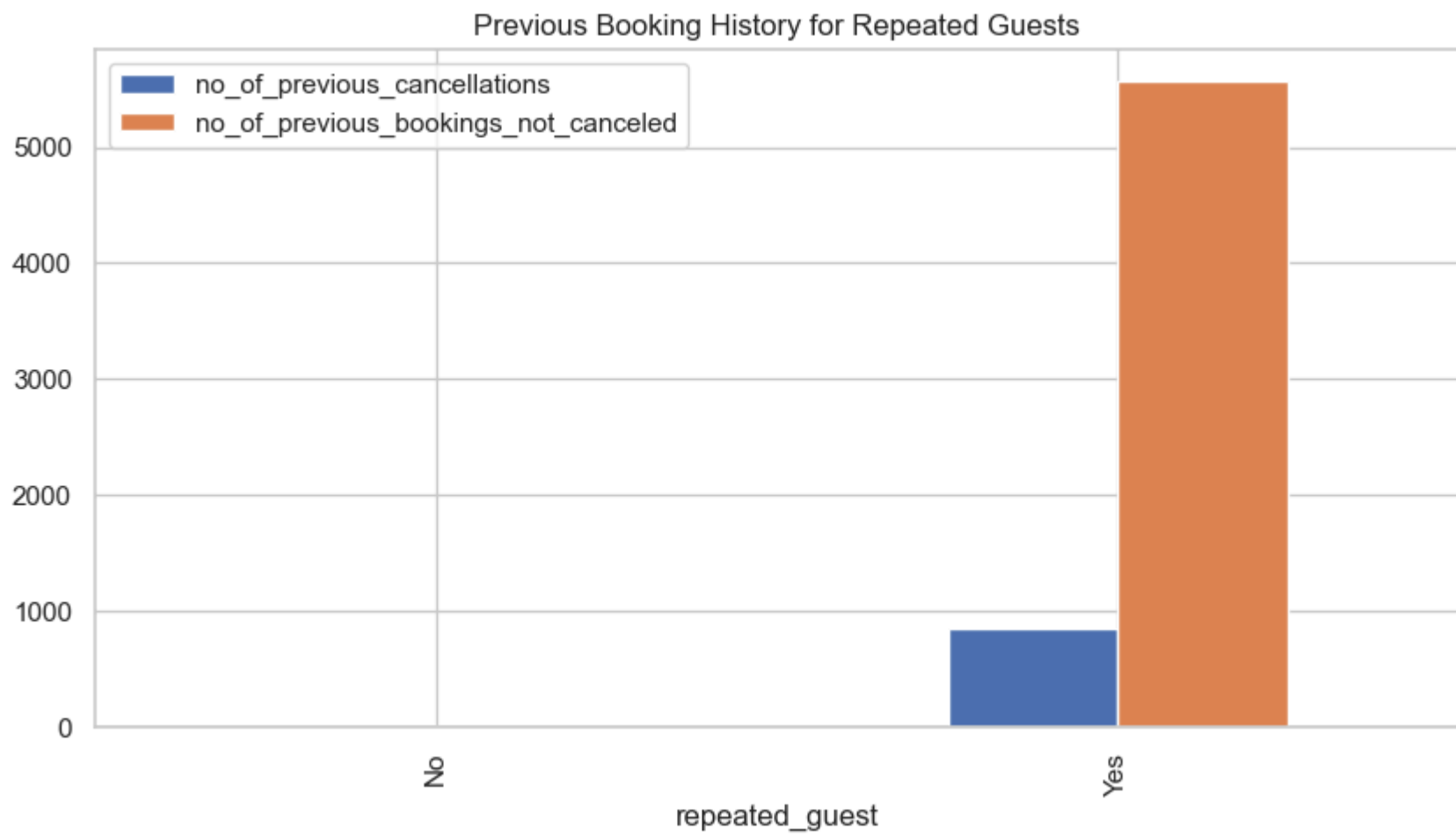
In [127…

```
plt.figure(figsize=(7,5))
sns.countplot(data=df, x='booking_status')
plt.title('Booking Status Overview')
plt.show()
```



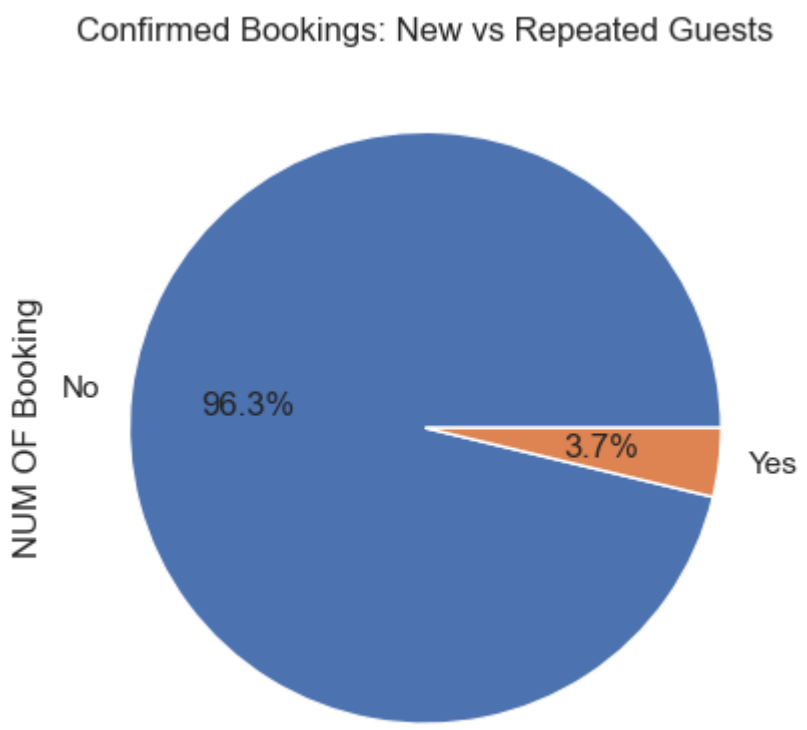
In [128…

```
guest_analysis = df.groupby('repeated_guest')[['no_of_previous_cancellations', 'no_of_previous_bookings_not_canceled']].sum()
guest_analysis.plot(kind='bar', figsize=(10,5))
plt.title('Previous Booking History for Repeated Guests')
plt.show()
```



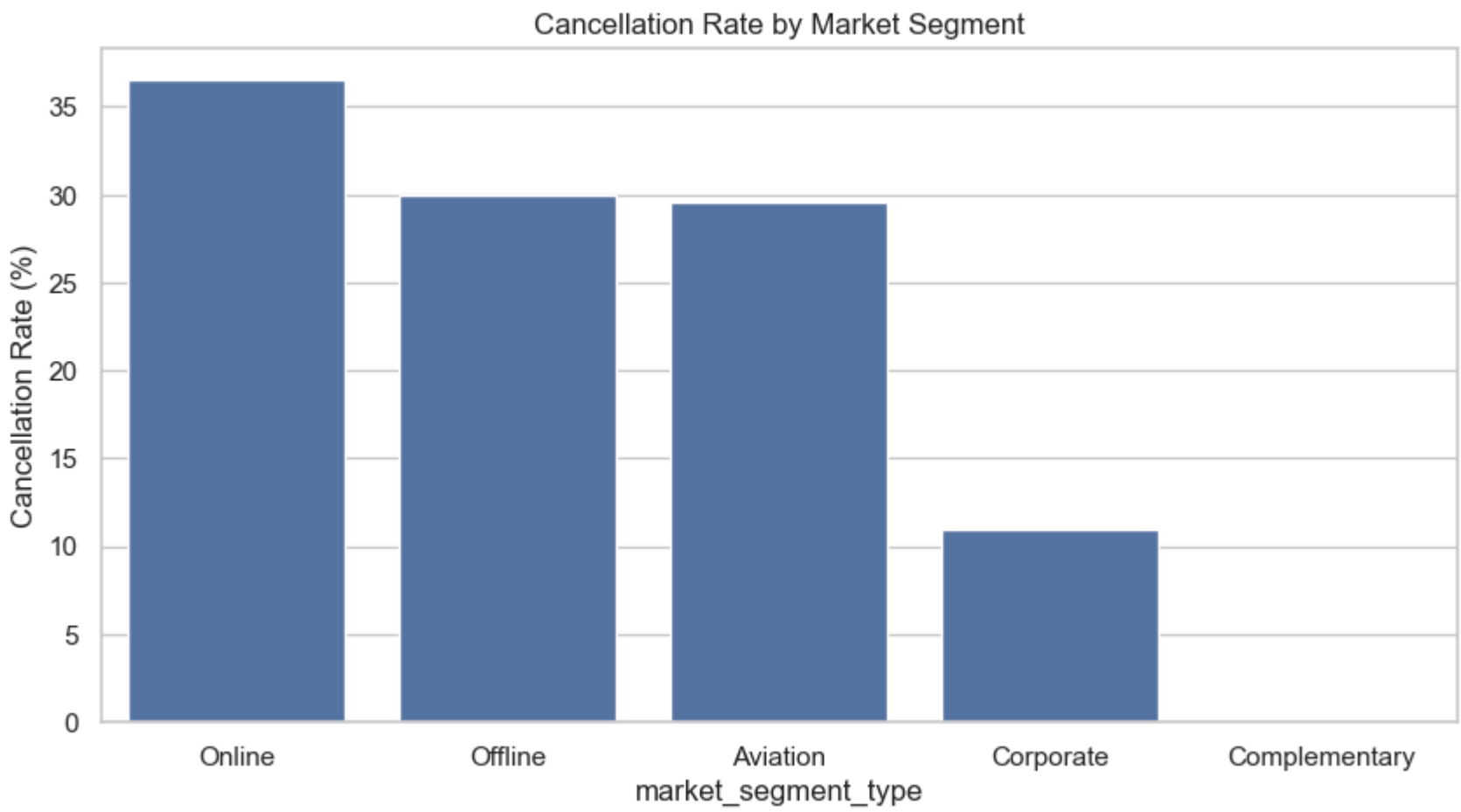
In [129…

```
loyalty_analysis['NUM OF Booking'].plot(kind='pie', autopct='%1.1f%%', title='Confirmed Bookings: New vs Repeated Guests')
plt.show()
```



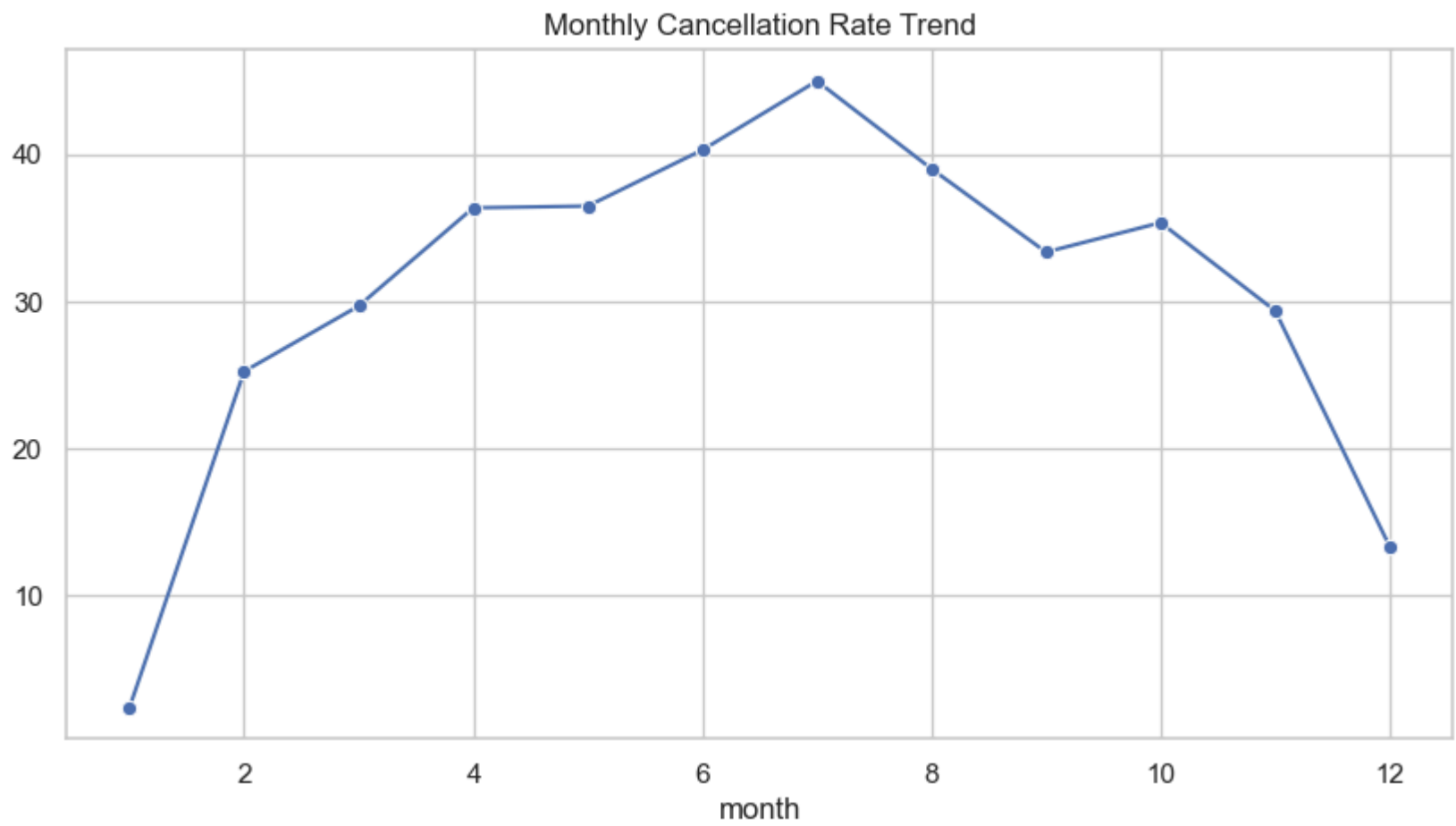
In [130…

```
cancel_rate = (df.groupby('market_segment_type')['is_canceled'].mean()*100).sort_values(ascending=False)
plt.figure(figsize=(10,5))
sns.barplot(x=cancel_rate.index, y=cancel_rate.values)
plt.ylabel('Cancellation Rate (%)')
plt.title('Cancellation Rate by Market Segment')
plt.show()
```



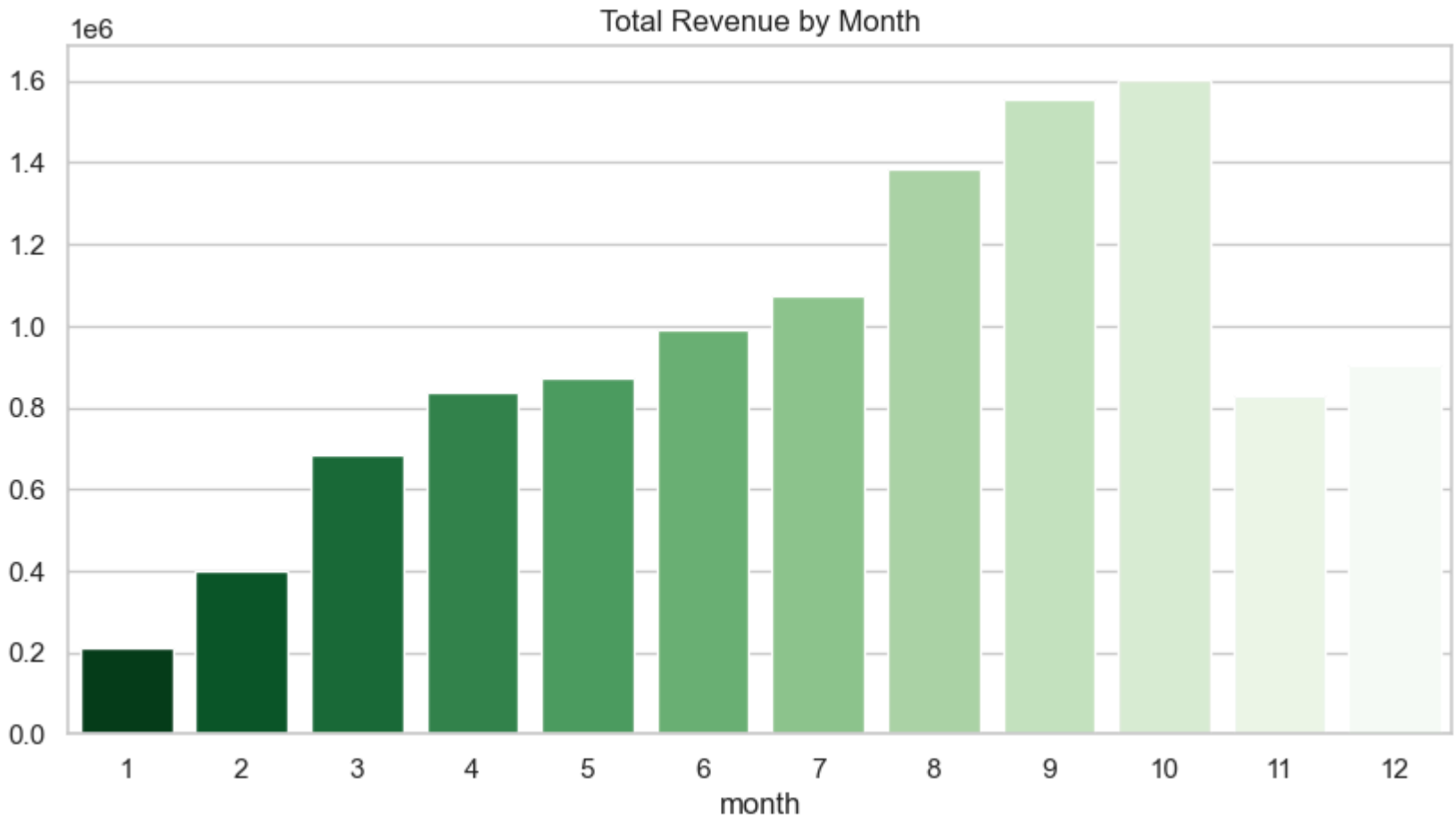
In [131…

```
month_cancel = (df.groupby('month')['is_canceled'].mean()*100)
plt.figure(figsize=(10,5))
sns.lineplot(x=month_cancel.index, y=month_cancel.values, marker='o')
plt.title('Monthly Cancellation Rate Trend')
plt.show()
```



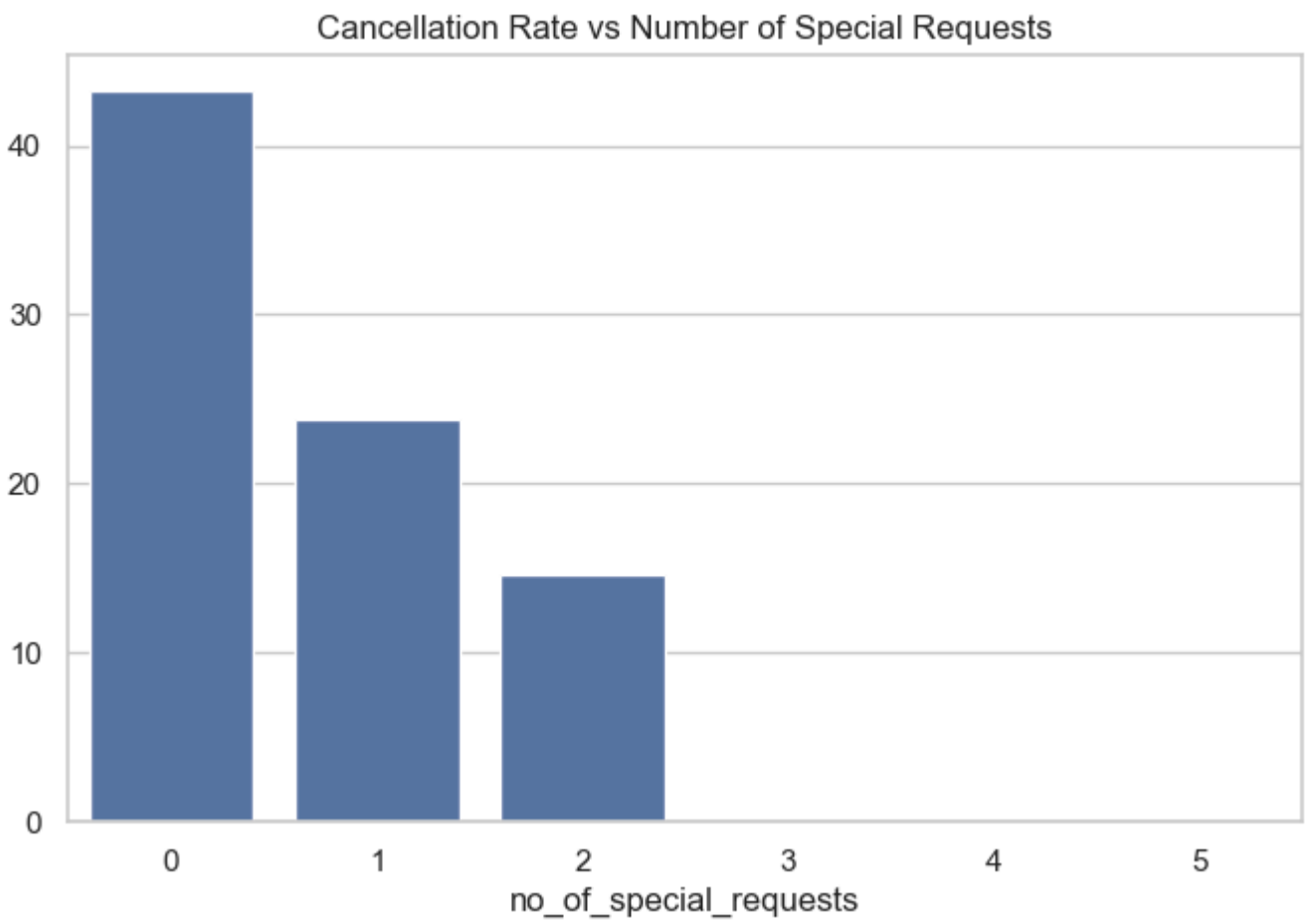
In [134…

```
month_rev = df.groupby('month')['revenue'].sum().sort_values(ascending=False)
plt.figure(figsize=(10,5))
sns.barplot(
    x=month_rev.index,
    y=month_rev.values,
    hue=month_rev.index,
    palette='Greens_r',
    legend=False
)
plt.title('Total Revenue by Month')
plt.show()
```



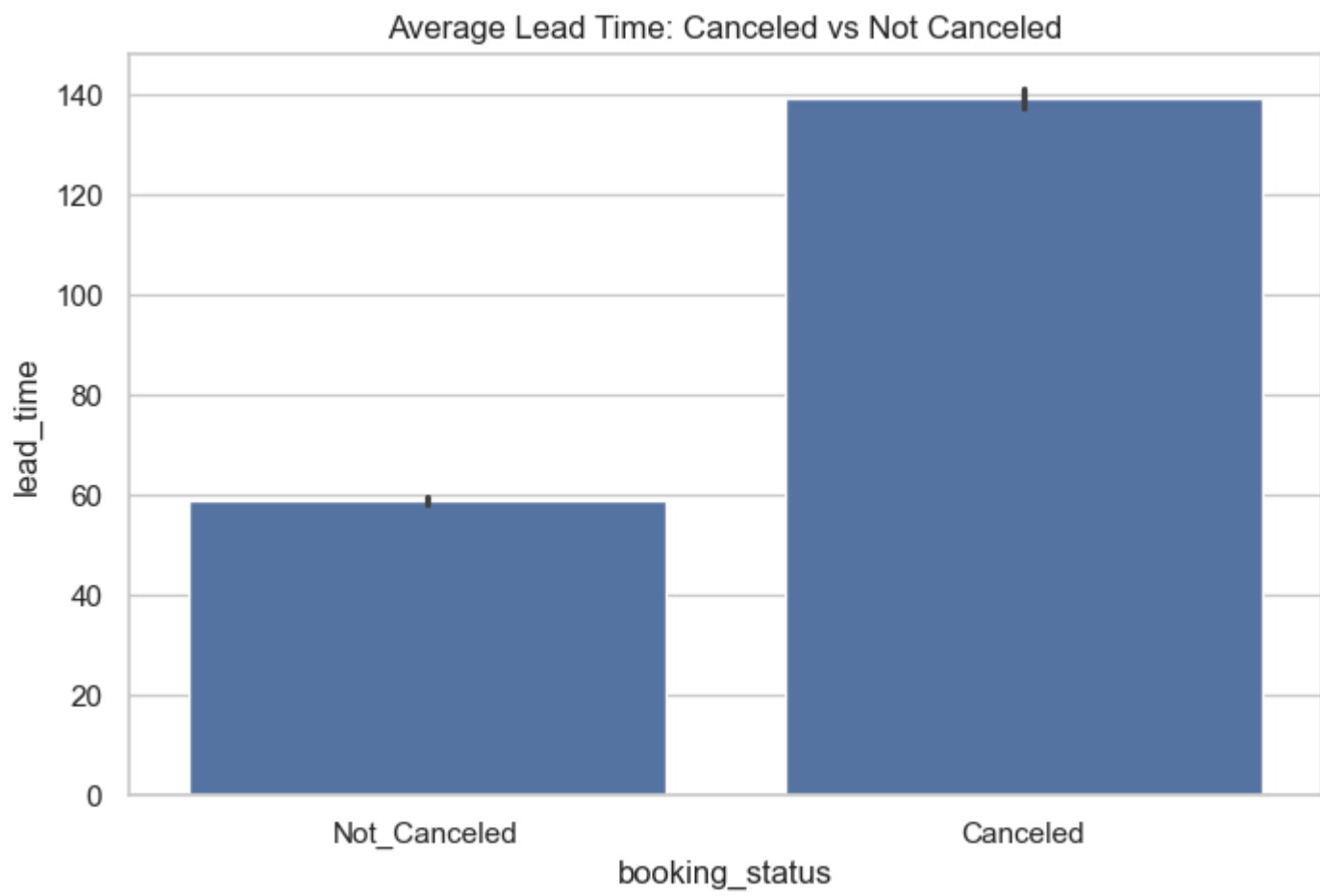
In [135…

```
spec_cancel = (df.groupby('no_of_special_requests')['is_canceled'].mean()*100)
plt.figure(figsize=(8,5))
sns.barplot(x=spec_cancel.index, y=spec_cancel.values)
plt.title('Cancellation Rate vs Number of Special Requests')
plt.show()
```



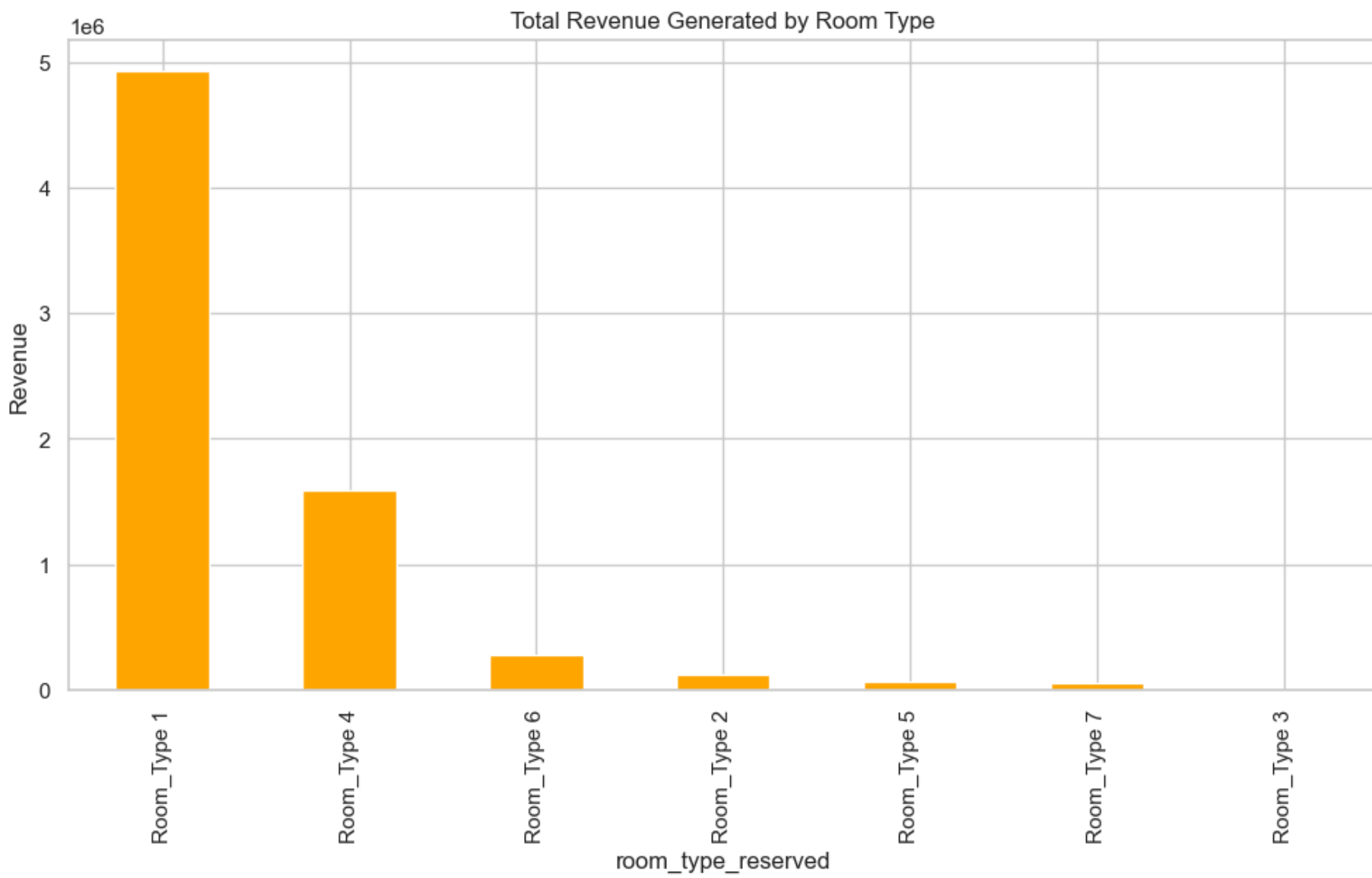
In [136…

```
plt.figure(figsize=(8,5))
sns.barplot(data=df, x='booking_status', y='lead_time', estimator='mean')
plt.title('Average Lead Time: Canceled vs Not Canceled')
plt.show()
```



In [137…

```
plt.figure(figsize=(12,6))
room_revenue.plot(kind='bar', color='orange')
plt.title('Total Revenue Generated by Room Type')
plt.ylabel('Revenue')
plt.show()
```



In [138…

```
df.to_csv('clean_data')
```

In [ ]: